

**ENVIRONMENTAL SCIENCE AND SUSTAINABLE
ENGINEERING**

TECHNICAL REPORT

**IMPROPER DISPOSAL OF ELECTRONIC
WASTE**

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1. ABSTRACT

Electronic waste, commonly known as e-waste, is one of the rapidly increasing waste problems associated with modern technology. Mobile phones, computers, laptops, chargers, batteries, televisions, printers and other electronic devices are frequently replaced due to technological advancement, damage or changing consumer preferences. When these unwanted devices are not managed properly, they may be mixed with general waste or disposed of through unsuitable methods.

Although people are increasingly aware that e-waste should be recycled responsibly, awareness alone may not always result in action. Many users may not be motivated to take the additional effort required to identify, collect and hand over their unwanted electronic items through appropriate channels. This creates a gap between environmental awareness and actual participation in responsible e-waste management.

This report proposes a solution called the **E-Waste Reward & Eco-Point System**. The proposed system encourages responsible e-waste disposal by providing users with digital eco-points when they participate in verified e-waste collection and recycling activities. These points can later be exchanged for suitable rewards, recognition, certificates or discounts provided by participating institutions or organizations.

The system combines technology with positive behavioural incentives. Users can register their e-waste submissions, participate in collection drives, receive verified eco-points and monitor their environmental contribution through a digital dashboard. Administrators can verify submissions, manage points, organize collection drives and monitor participation.

The proposed solution aims to increase participation in responsible e-waste management while creating awareness about recycling, reuse and resource recovery. It can initially be implemented in colleges, schools, residential communities or organizations and can later be expanded to larger communities.

2. INTRODUCTION

Electronic devices have become an essential part of everyday life. Smartphones, laptops, computers, tablets, televisions, printers, headphones, chargers and other electrical products are widely used in homes, educational institutions and workplaces. The rapid development of technology has also resulted in people replacing electronic devices more frequently.

When electronic products are no longer wanted or usable, they become electronic waste. E-waste is different from ordinary household waste because electronic products contain a mixture of useful materials and components that require appropriate handling. Materials such as metals and other recoverable components can potentially be reused or recycled when e-waste is collected and processed properly.

Improper disposal of e-waste can occur when electronic devices are thrown into general waste, dumped in unsuitable locations, burned or handed over through unreliable channels. Such practices can contribute to environmental pollution and the loss of valuable resources.

One of the important challenges in e-waste management is **user participation**. Even when people know that electronic waste should be recycled, they may postpone disposal, store old devices at home or simply discard them with other waste because responsible disposal may seem inconvenient.

Therefore, technology can play an important role in encouraging better disposal behaviour. Instead of providing only information, a digital system can actively motivate people by rewarding responsible environmental actions.

The **E-Waste Reward & Eco-Point System** is proposed as a solution to this behavioural challenge. It uses a points-based incentive system to encourage users to participate in verified e-waste collection and recycling activities.

3. PROBLEM STATEMENT

Problem: IMPROPER DISPOSAL OF ELECTRONIC WASTE

Electronic waste refers to discarded electrical and electronic equipment such as mobile phones, computers, laptops, televisions, chargers, batteries, printers and other electronic devices. The rapid growth in electronic consumption and frequent replacement of older products have contributed to the increasing generation of e-waste.

Improper disposal occurs when unwanted electronic products are mixed with general municipal waste, dumped in unsuitable locations, burned, dismantled through unsafe practices or handed over through unverified channels. Such practices can result in the loss of valuable recoverable materials and may increase environmental risks.

Electronic waste requires appropriate collection, segregation, transportation and treatment. When these processes are not properly managed, electronic components may not be recovered efficiently and may instead become part of inappropriate waste streams.

Another major issue is the lack of motivation and participation in responsible e-waste disposal. People may know that e-waste should be recycled but may not act because collection facilities are inconvenient, information is difficult to find, or there is no direct incentive to participate.

Therefore, the environmental problem addressed in this project is the improper disposal and inadequate management of electronic waste, resulting in environmental risks, loss of recoverable resources and low participation in responsible e-waste collection and recycling.

4. SUSTAINABILITY ANALYSIS

The proposed project contributes to environmental sustainability by encouraging people to participate in responsible e-waste collection and recycling. Instead of treating unwanted electronic products as ordinary waste, the system promotes appropriate end-of-life management and resource recovery.

4.1 SUSTAINABLE DEVELOPMENT GOALS

The proposed system is connected to several United Nations Sustainable Development Goals.

SDG 12 – RESPONSIBLE CONSUMPTION AND PRODUCTION

The project directly supports responsible management of products at the end of their useful life. By encouraging users to return unwanted electronic products through proper channels, the system promotes recycling, reuse and resource recovery.

SDG 11 – SUSTAINABLE CITIES AND COMMUNITIES

Proper management of electronic waste contributes to cleaner communities and more organized waste-management practices. The system can support community-level collection drives and responsible disposal campaigns.

SDG 13 – CLIMATE ACTION

Responsible recycling and recovery of materials can reduce the need for extraction and processing of new raw materials. Therefore, improved e-waste management can indirectly contribute to reducing environmental impacts.

SDG 3 – GOOD HEALTH AND WELL-BEING

Proper collection and treatment of e-waste can help reduce risks associated with unsafe handling and disposal practices.

4.2 ISO 14001

ISO 14001 provides a framework for environmental management systems. The proposed system can support environmental management by maintaining digital records of e-waste collection, participation and environmental activities. These records can help organizations monitor their waste-management performance and identify areas for improvement.

5. PROPOSED ENGINEERING / IT SOLUTION

Solution: E-WASTE REWARD & ECO-POINT SYSTEM

The proposed solution is an **E-Waste Reward & Eco-Point System**, a digital platform designed to encourage people to dispose of electronic waste responsibly. Instead of simply providing a platform for collecting e-waste, the system focuses on motivating users through a reward-based approach. Users can register on the platform and submit details about the electronic items they wish to dispose of, such as old mobile phones, chargers, keyboards, laptops, printers, and other electronic devices. After the submission is verified by an administrator or collection coordinator, the user receives a specific number of eco-points based on the type and quantity of e-waste handed over.

The accumulated eco-points can then be exchanged for suitable rewards such as digital certificates, eco-badges, campus recognition, event benefits, or discounts from participating partners. A user dashboard can allow individuals to view their points, previous submissions, redeemed rewards, and overall contribution to e-waste management. The system can also include a leaderboard to encourage friendly participation among students or members of a community.

The administrator can manage user registrations, verify e-waste submissions, approve eco-points, organize collection drives, manage available rewards, and generate reports. The system can be developed as a web application using technologies such as HTML, CSS, JavaScript, a backend framework like Node.js or Python, and a database such as MySQL or PostgreSQL. By combining digital technology with incentives, the system aims to make responsible e-waste disposal more convenient, engaging, and rewarding for users.

6. FEASIBILITY ANALYSIS

The E-Waste Reward & Eco-Point System is technically and practically feasible as a student engineering project. The basic version can be developed using commonly available software technologies without requiring expensive hardware.

6.1 TECHNICAL FEASIBILITY- The system can be developed using standard web-development technologies. User registration, databases, point calculations, dashboards and reward management are all technically achievable.

6.2 ECONOMIC FEASIBILITY- A basic prototype can be developed using free and open-source software. The reward system does not necessarily require expensive physical rewards. Digital certificates, badges, recognition and institutional benefits can be used.

6.3 OPERATIONAL FEASIBILITY- The system can be operated by a college administrator, environmental club or authorized collection coordinator. The administrator can verify submissions and manage eco-points.

6.4 USER FEASIBILITY- The system is designed to be simple for users. This can encourage greater participation of users.

SUBMIT → VERIFY → EARN → REDEEM

6.5 SCALABILITY- The system can initially be implemented within a college campus and can later be expanded to multiple colleges, residential communities, offices, schools, and local governments.

6.6 LIMITATIONS- The system may face certain limitations such as false or duplicate submissions, the need for proper verification, limited availability of collection partners, the cost of providing rewards, maintenance of user and collection data, and the requirement for regular e-waste collection drives. These issues can be addressed through administrator verification, unique submission IDs, predefined point rules, and regular auditing.

7. EXPECTED ENVIRONMENTAL BENEFITS

The main expected environmental benefit is an increase in responsible e-waste collection and recycling.

EXPECTED BENEFIT	HOW THE SYSTEM HELPS
INCREASED E-WASTE COLLECTION	REWARDS USERS FOR VERIFIED E-WASTE HANDOVER
REDUCED IMPROPER DISPOSAL	ENCOURAGES USERS TO USE RESPONSIBLE DISPOSAL CHANNELS
RESOURCE RECOVERY	SUPPORTS THE FLOW OF E-WASTE TOWARDS APPROPRIATE RECYCLING
INCREASED AWARENESS	DIGITAL REWARDS AND CAMPAIGNS KEEP USERS ENGAGED
GREATER COMMUNITY PARTICIPATION	COLLECTION DRIVES CAN USE ECO-POINTS TO MOTIVATE PARTICIPATION
BETTER MONITORING	DIGITAL RECORDS HELP TRACK COLLECTION ACTIVITIES

The system can encourage environmental responsibility among students and community members. Tracking points and participation history can make responsible e-waste disposal more engaging and measurable.

For example, a college can conduct an “**E-Waste Collection Week**”, where students receive eco-points for handing over eligible e-waste. A leaderboard can also encourage friendly participation among classes or departments.

The system can be evaluated using indicators such as the **total e-waste collected, number of participating users, number of verified submissions, number of collection drives, amount of e-waste sent for recycling, and number of rewards redeemed.**

8. JUSTIFICATION OF THE PROPOSED SOLUTION

The proposed E-Waste Reward & Eco-Point System is justified because improper disposal of electronic waste is not only caused by a lack of collection facilities but also by low awareness, inconvenience, and lack of motivation among users. Many people may keep old electronic devices at home or dispose of them along with regular waste because they do not see any immediate benefit in recycling them.

The reward-based approach provides users with a simple motivation to participate in responsible e-waste disposal. By awarding eco-points for verified submissions, the system makes environmentally responsible actions more attractive and measurable. Users can track their points and participation, which can encourage them to continue taking part in future collection activities.

The system is particularly suitable for colleges and educational institutions because students can participate through collection drives, class competitions, leaderboards, certificates, and other campus-based rewards. It can also create awareness and encourage students to influence responsible disposal practices within their families and communities.

Another advantage is that the system does not require expensive hardware for its basic implementation. It can be developed using commonly available web technologies and can use digital rewards or recognition, reducing the overall operating cost. The system can also maintain digital records of e-waste submissions, making it easier to monitor participation and measure environmental performance.

Unlike a conventional e-waste collection portal, the focus of this solution is **behavioural motivation through rewards**. Therefore, it provides an additional approach to increasing participation in proper e-waste disposal while supporting sustainable waste management and resource recovery.

9. CONCLUSION

Improper disposal of electronic waste is an increasing environmental concern due to the rapid growth in the use and replacement of electronic devices. When e-waste is not handled properly, valuable materials may be lost and environmental risks can increase. Therefore, encouraging people to participate in responsible e-waste management is an important part of addressing the problem.

The proposed **E-Waste Reward & Eco-Point System** provides a practical IT-based approach to this issue. Instead of focusing only on collecting e-waste, the system motivates users by providing eco-points for verified responsible disposal. These points can be exchanged for rewards, certificates, recognition, or other suitable benefits. This makes responsible disposal more engaging and encourages users to participate regularly.

The system can also help colleges and communities organize collection drives and monitor participation through digital records. Features such as dashboards, leaderboards, and collection statistics can increase awareness and encourage friendly competition among participants.

Overall, the proposed system combines **technology, environmental sustainability, and positive incentives** to encourage better e-waste disposal practices. It can initially be implemented on a small scale, such as within a college campus, and later expanded to schools, communities, offices, and other organizations. Its success can be measured through factors such as the amount of e-waste collected, number of participants, verified submissions, and recycling activities. Thus, the system can contribute to reducing improper e-waste disposal while promoting a more responsible and sustainable approach to electronic waste management.

10. REFERENCES

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